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Review

Digital Adolescent Sexual and Reproductive Health in Low- and Middle-Income Countries: A Scoping Review

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Abstract: Every adolescent should have the right to make their own choices about their sexual and reproductive health (SRH). Achieving this goal can be challenging, especially in low- and middle-income countries (LMICs), where social norms and insufficient resources may limit access to information. Technology is increasingly being leveraged in LMICs to reach adolescents. We searched the literature to understand the landscape of digital SRH interventions in LMICs. The review addressed three questions: (1) What are the interventions and how effective are they? (2) What are adolescent preferences for information? And (3) What do the interventions cost? There is a wide variety in the populations addressed (e.g., adolescents with HIV, pregnant/post-partum adolescents) and the locations of the interventions, with the majority in sub-Saharan Africa. The types of interventions were evenly distributed between "push" approaches, where adolescents are sent information; "pull" approaches, where adolescents reach information from a website or app; and "two-way" approaches, which combine push and pull aspects. The most positive features identified were convenience and privacy. The major challenge is inadequate smartphone penetration in very low-income adolescents, especially girls. The evidence on the effectiveness of these interventions is mixed. Most studies show that SRH knowledge increases, but this does not necessarily lead to behavior change.

Keywords: adolescents; teens; sexual and reproductive health; pregnancy; mobile technology; cell phones; low- and middle-income countries

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1. Introduction

Adolescence and young adulthood are critical periods characterized by significant biological, psychological, and social transformations, including the development of sexual and reproductive health (SRH). The United Nations Population Fund (UNFPA) defines good SRH as a state of complete physical, mental, and social well-being in all matters relating to sexual and reproductive health, and implies that people are able to have a satisfying and safe sex life. This holistic perspective underscores the necessity for adolescents to access accurate information on safe, effective, affordable, and acceptable contraceptive methods and to be empowered to protect themselves from sexually transmitted infections (STIs) and to make informed choices about their SRH (Sexual & Reproductive Health, n.d.).

Despite global advancements, achieving optimal SRH remains a formidable challenge in low- and middle-income countries (LMICs) due to pervasive social norms, economic barriers, and infrastructural deficiencies that restrict access to SRH information and services (Ames et al., 2019; Batista et al., 2020; Goldstein et al., 2023; Gonsalves et al., 2019; Ippoliti & L'Engle, 2017). Adolescents in these regions often lack reliable sources of SRH

information, contributing to adverse health outcomes such as unintended pregnancies, STIs, and unsafe abortions.

However, the rapid proliferation of mobile technology presents a transformative potential for SRH interventions. Mobile phones, which have become ubiquitous even in resource-limited settings, offer a promising platform for delivering SRH information and services (Feroz et al., 2021; K. L. L'Engle et al., 2016; Mangone et al., 2016; Palmer et al., 2020). The penetration of mobile technology in LMICs has catalyzed innovative initiatives aimed at leveraging digital tools to reach adolescents with crucial SRH information. These initiatives encompass a range of modalities, including mobile health (mHealth) applications, text-based interventions, social media platforms, and telemedicine services (Reynolds et al., 2019; Somefun et al., 2021).

Existing interventions include push notifications and other texts or reminders sent directly to mobile phones, information that can be sought out by users on mobile applications or websites, and a combination of the two approaches. Most interventions follow one of two models: (1) provide information about SRH or telehealth services fully remotely to help reach populations who might not otherwise have access to trusted advice or care; or (2) supplement in-person care with virtual, often automated reminders and information to keep users engaged in between visits. An example of model one might include details about where users can access no or low-cost contraceptives, while an example of model two might include automated reminders for users taking pre-exposure prophylaxis to prevent HIV.

In anticipation of commissioning work to support technological innovations providing SRH information for adolescents in Brazil, Guinea, and India, ChildFund International, a health and wellness nonprofit organization based in the United States, undertook this comprehensive review to evaluate the existing landscape of technology-enabled SRH programs targeting adolescents in LMICs. This review is important and necessary for the following several reasons: (1) a due diligence process was needed before conceptualizing or designing any digital health interventions to better understand what did and did not work as well as best practices in the field; (2) scalability and cost-effectiveness were also important to understand before testing or implementing any digital health programs; and (3) adolescents' preferences needed to be understood to ensure acceptability and reduce the likelihood of attrition.

This review is guided by three primary research questions:

1. Scaling and Effectiveness of Technology-Enabled Interventions: What technology-enabled adolescent SRH behavior change interventions have been scaled in LMICs, and what evidence exists regarding their effectiveness?
2. Adolescents' Preferences for Digital Program Delivery: What does the literature reveal about adolescents' preferences and engagement with digital SRH programs?
3. Investment and Development of Digital SRH Interventions: What insights can be gained regarding the level and types of investments organizations make to develop and deliver digital adolescent SRH behavior change interventions?

This review examines the current state of technology-enabled SRH interventions for adolescents in LMICs. By synthesizing existing evidence, this review aims to inform future program design and implementation, identify gaps in the current knowledge base, and recommend areas for further research and innovation in digital SRH interventions.

2. Materials and Methods

2.1. Literature Search Strategy and Inclusion Criteria

To comprehensively review the current landscape of technology-enabled sexual and reproductive health interventions for adolescents in low- and middle-income countries,

a systematic literature search was conducted. The inclusion criteria were meticulously defined to ensure the relevance and quality of the selected studies.

- Types of Evaluations: We included studies featuring process evaluations, quasi-experimental and experimental evaluations, as well as cost or cost-effectiveness evaluations.
- Case Studies: Case studies were included if they provided substantial qualitative or quantitative information to answer the research questions.
- Literature Types: Both peer-reviewed and non-peer-reviewed literature, such as reports and briefs, were considered.
- Study Designs: Qualitative, quantitative, and mixed-methods studies were included to ensure a comprehensive understanding of the topic.
- Publication Date: Only studies published from 2010 to 2023 were included to reflect the most recent developments in technology and SRH.
- Geographic Scope: The focus was on interventions conducted in LMICs as classified by the World Bank (World Bank Open Data, n.d.).
- Participant Age Range: Studies involving participants aged 10–24 years were included to cover the adolescent and young adult population.
- Intervention Type: We included technology-enabled interventions utilizing cell phones, computers, and tablets but excluded those relying on mass media communications such as radio or television.
- Language: Only English-language papers were included to maintain consistency in data interpretation.
- Exclusions: Protocol papers (those outlining research plans without results), dissertations, and theses were excluded.

2.2. Search Keywords and Databases

To capture relevant studies, we used a set of predefined keywords across several domains, which were searched in PubMed and Google Scholar in February 2023. PubMed was chosen as the first database because it is listed by the NIH Library as the premier healthcare and public health database. Google Scholar was chosen as a second database due to the size and scope of the papers that are included in the database. The following keywords were included:

- Age Range Keywords: Adolescent, Youth, Teen, Young Adult.
- Topic Keywords: Sexual and Reproductive Health, Family Planning, Contraception, Sexually Transmitted Disease, STD, SRH, HIV, AIDS, Health Education, Health Information, Female Genital Mutilation, FGM, Female Circumcision.
- Location Keywords: Developing Country, Global South, Third World, Less Developed Country, Low Income Country, Middle Income Country, Brazil, Guinea, India, Africa, Latin America, South America, Central America, Pakistan, Southeast Asia, China.
- Technology Keywords: Technology, Cell Phone, Cellular, Tablet, Computer, Online, mHealth.

2.3. Study Selection and Screening Process

The initial search yielded 329 papers after removing 20 duplicates. An additional 10 papers were identified through manual reference list scanning, resulting in a total of 339 papers. The selection process involved three stages:

1. Title and Abstract Screening: Titles and abstracts were reviewed for relevance based on the inclusion criteria. Studies that did not meet the criteria were excluded.

2. Full-Text Review: Full texts of the remaining papers were thoroughly reviewed to confirm eligibility. Studies that failed to meet the inclusion criteria upon full-text review were excluded.
3. Data Extraction: Relevant data were systematically extracted from the included studies, focusing on study design, intervention details, outcomes, and key findings.

The flow of studies through the review process is detailed in Figure 1. This methodology complies with the PRISMA checklist for scoping reviews.

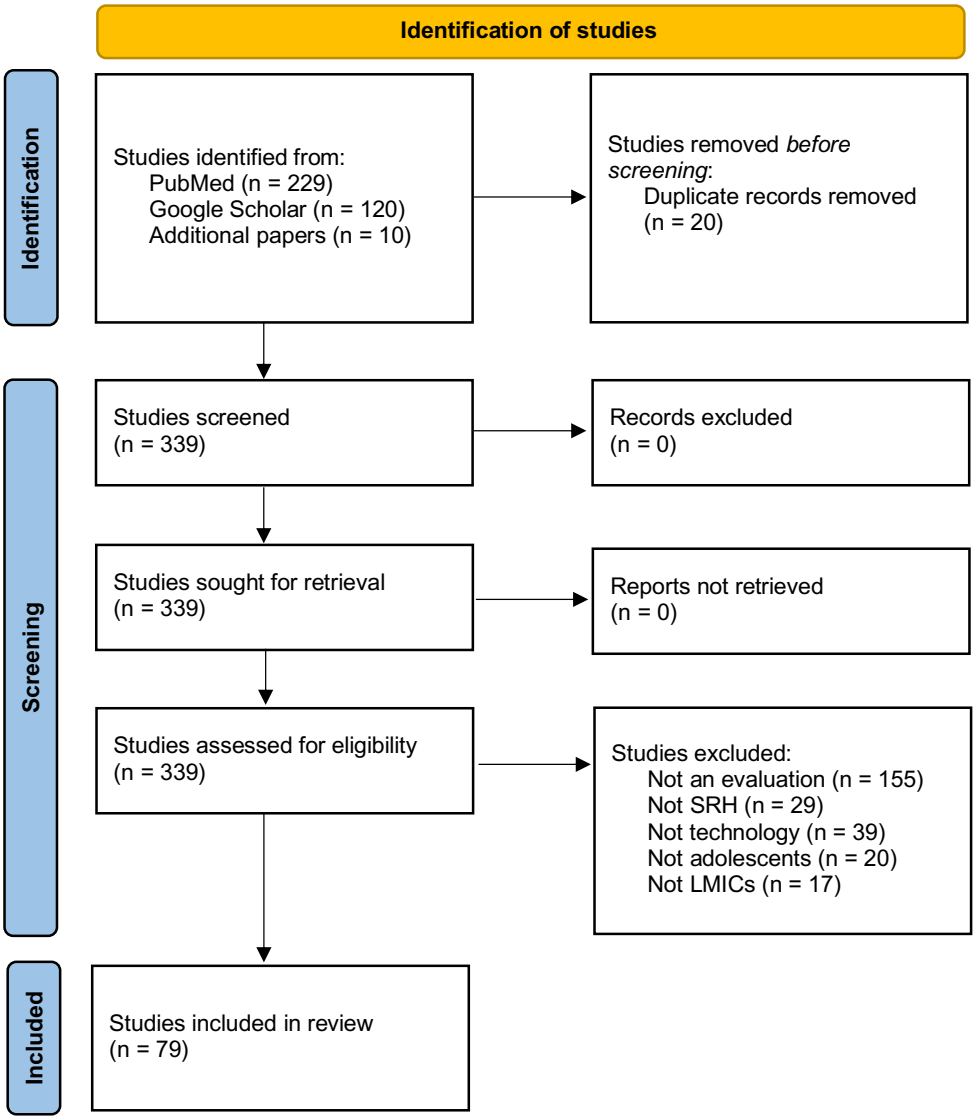


Figure 1. Paper inclusion flowchart.

2.4. Analysis

Out of the initial 339 papers, 79 met the inclusion criteria and were included in the final review. Approximately half the papers (45.6%) were published from 2020 to 2023, with the remainder in the 2010s. Almost half the papers concerned either HIV (17 papers) or sexual and reproductive health (18 papers). The remainder targeted a general population of adolescents. Well over half of studies (50 papers) took place in Africa, with the others primarily in Latin America and Asia.

Once the papers were identified, we reviewed each paper to determine whether it was a qualitative study or a quantitative study as well as the relative quality of this study. We reviewed the methods of each, according to accepted criteria for the type of study. For

example, for qualitative studies, we analyzed how interview or focus group participants were selected and whether protocols were adequate. For quantitative studies, we assessed generalizability, adequate sample size, and the type of data collected. We then examined the results of the studies, in terms of the degree to which they answered the three research questions for the review.

This methodological approach provides a detailed and nuanced understanding of the effectiveness, preferences, and investment strategies associated with technology-enabled SRH interventions for adolescents in LMICs. The findings from this review will inform future program design and implementation, identify gaps in the current knowledge base, and recommend areas for further research and innovation in digital SRH interventions.

3. Results

3.1. Types of Studies Analyzed

Qualitative studies comprised approximately one-third of the included studies, employing methods such as focus groups, structured key informant interviews, and participant observation. These studies provided in-depth insights into adolescents' experiences and preferences regarding digital SRH interventions.

Quantitative data were derived from 12 surveys, which provided statistical information on adolescent ownership and use of cell phones, as well as their engagement with digital SRH programs. These surveys were instrumental in addressing the first research question regarding technological innovations and the second research question on adolescent preferences.

Additional insights were gathered from six cross-cutting case studies and an inventory of sexual and reproductive health apps available on Apple and Android app stores. These studies contributed to a comprehensive understanding of the types of interventions and their acceptability.

To assess the impact of the interventions, we reviewed 12 systematic reviews and 16 impact studies. The systematic reviews included several high-quality reviews utilizing Cochrane methods, with two incorporating meta-analyses. The impact studies, primarily small randomized clinical trials, often faced limitations due to small sample sizes, leading to null findings. Despite this, these studies provided valuable information on intervention types and their acceptability.

The review identified a limited number of studies addressing cost-related aspects of digital SRH interventions. Of the 10 papers discussing cost, most focused on the cost to adolescents rather than the implementation costs. Only three papers provided detailed information on the cost to implementers, highlighting a gap in the literature regarding the financial sustainability of these interventions.

3.2. Types of Interventions Analyzed

The reviewed interventions were categorized into three types: "push" interventions, "pull" interventions, and "two-way" interventions. Push interventions involve sending reminders or SRH information directly to adolescents via text messages. The review identified 11 push interventions. For instance, the "Tu Decides" program in Bolivia targets older female adolescents by sending up to three text messages daily about contraceptive options (McCarthy et al., 2020). Similarly, the "SITA" program in Uganda sends messages to adolescents in HIV treatment about their adherence rates, with variations showing individual versus group adherence (MacCarthy et al., 2020). The "MOTIF" program in Cambodia (Brody et al., 2022) provides post-abortion text messages about contraceptive options, and a program in Zimbabwe (Dhakwa et al., 2021) sends reminders to adolescents to attend referred appointments for sexual and reproductive health and HIV services.

Pull interventions provide tools for adolescents to seek SRH information independently, such as apps or internet access. The review identified 11 pull interventions. An example is "PeerNaija" in Nigeria, a computer game developed with adolescent input to educate about SRH (Ahonkhai et al., 2021). The "ARMADILLO" program in several African countries offers on-demand access to SRH information, complemented by reminder text messages (Gonsalves et al., 2019). Another notable example is an internet-based SRH education program from China, featuring eight structured SRH sessions (Hu et al., 2023). These interventions empower adolescents to actively seek information, promoting engagement and self-efficacy.

The review identified 13 two-way interventions, reflecting an increasing trend towards fostering interactive communication in SRH programs. These programs combine information delivery with interaction, demonstrating greater success. For instance, "Khuluma" in South Africa uses facilitator-enabled group conversations for youth with HIV (Atujuna et al., 2021). In Kenya, the "ELIMIKA" web-based peer support platform includes blog posts by project coordinators, a Q&A section for health providers, and private messaging for confidential communication (Ivanova et al., 2019). Another Kenyan program overcomes the need for smartphones by using USSD (Unstructured Supplementary Service Data), enabling interactive SRH programming via regular cell phones (Karusala et al., 2021).

The evolution of technology, particularly the increased availability of smartphones, has enhanced the feasibility of interactive and internet-based interventions. This review underscores the diversity and effectiveness of these technology-enabled SRH interventions for adolescents in LMICs. While push interventions deliver crucial information directly, pull interventions encourage active information-seeking, and two-way interventions foster meaningful interactions, addressing various aspects of adolescent SRH needs. Table 1 details the number of interventions and associated citations for each type. The number of interventions classified in Table 1 (35) does not match the number of papers reviewed, because often a particular intervention can be identified in more than one paper, either because an evaluator published multiple papers on a single intervention or because an intervention is included in multiple review papers.

Table 1. Types of SRH technology interventions in LMICs.

Type of Intervention	Number of Interventions	Citations
Push	11	See: (Brody et al., 2022; Dhakwa et al., 2021; Fatori et al., 2020; Jamison et al., 2013; MacCarthy et al., 2020; McCarthy et al., 2020; Rodrigues et al., 2014; Rokicki et al., 2017; Smith et al., 2015; Unger et al., 2018; Waldman & Stevens, 2015)
Pull	11	See: (Ames et al., 2019; Dev et al., 2019; Gichangi et al., 2022; Hernández-Torres et al., 2022; Hu et al., 2023; K. L'Engle et al., 2017; Lopez et al., 2014; Nuwamanya et al., 2020; Sabben et al., 2019; Venter et al., 2019; Ybarra et al., 2012)
Two-way	13	See: (Ahonkhai et al., 2021; Atujuna et al., 2021; Batista et al., 2020; Bergam et al., 2022; Hacking et al., 2019; Hill et al., 2020; Ivanova et al., 2019; Karusala et al., 2021; Macharia et al., 2021; Rokicki et al., 2017; Thompson et al., 2019; Waldman & Stevens, 2015; Wang et al., 2022)
Total	35	

3.3. Effectiveness (Research Question One)

The evidence surrounding the effectiveness of the aforementioned technology-enabled adolescent SRH behavior change interventions in LMICs is mixed and characterized by a lack of methodologically rigorous studies. This conclusion is based not on the absence of excellent programs but on the challenges associated with designing and executing experimental studies of effectiveness. These studies are often difficult and expensive to conduct, and many existing studies suffer from flaws and biases, such as small sample sizes, selection issues, low response rates, high dropout rates, and a lack of credible control groups. Additionally, the rapidly evolving nature of technology interventions and the variability in types of interventions and target populations complicate the generalizability of results. For example, findings from a program implemented five years ago may not apply to current interventions, and results from a program for adolescents living with HIV in Africa may not be applicable to pregnant adolescents in Latin America.

Despite these challenges, some useful information has emerged from high-quality reviews and impact studies. Table 2 summarizes findings from 12 systematic reviews and 16 impact studies, revealing decidedly mixed results about the impact of SRH programs for adolescents in LMICs. This inconsistency aligns with the conclusions of many researchers who conducted high-quality reviews (Goldstein et al., 2023; Ippoliti & L'Engle, 2017; Lee et al., 2016; K. L. L'Engle et al., 2016; Nigenda et al., 2016; Portela et al., 2017). Substantial evidence indicates that many interventions have increased SRH knowledge among adolescents. Specifically, six studies documented improved SRH knowledge (Feroz et al., 2021; Hernández-Torres et al., 2022; Hu et al., 2023; Nuwamanya et al., 2020; Palmer et al., 2020; Sharma et al., 2022); another review of six studies found increased knowledge in two of them (Ivanova et al., 2019); one study documented increased access to contraceptive information (Feroz et al., 2021); and another found increased contraceptive knowledge (Riley, 2014). However, two studies reported no significant change in SRH knowledge (Jamison et al., 2013; Winskell et al., 2019), and another found no change in contraceptive knowledge (Gichangi et al., 2022).

Table 2. Review of impact evidence by type of impact and citations.

Type of Impact	Finding	Citations	Number of Citations
Knowledge Outcomes			
Knowledge of SRH	Significantly improved	See: (Feroz et al., 2021; Hernández-Torres et al., 2022; Hu et al., 2023; Nuwamanya et al., 2020; Palmer et al., 2020; Sharma et al., 2022)	6
Knowledge of SRH	Mixed: 2 of 17 HIV knowledge items improved	See: (Ivanova et al., 2019)	13
Knowledge of SRH	No significant change	See: (Jamison et al., 2013; Winskell et al., 2019)	2
Access to contraceptive information	Significantly improved	See: (Feroz et al., 2021)	1
Knowledge of contraception	Significantly improved	See: (Riley, 2014)	1
Knowledge of contraception	No significant change	See: (Gichangi et al., 2022)	1

Table 2. *Cont.*

Type of Impact	Finding	Citations	Number of Citations
Health Behaviors			
Risky sexual behavior	Significantly reduced	See: (Hernández-Torres et al., 2022)	1
Risky sexual behavior	No significant change	See: (Hu et al., 2023; Jamison et al., 2013; Lopez et al., 2014)	3
Condom use	Significantly increased	See: (Nuwamanya et al., 2020)	1
Condom use	No significant change	See: (Brody et al., 2022; Ybarra et al., 2013)	2
Contraceptive use	Significantly increased	See: (Nuwamanya et al., 2020; Palmer et al., 2020; Smith et al., 2015; Unger et al., 2018)	4
Contraceptive use	No significant change	See: (Riley, 2014)	1
Health Services			
Completed referral to sexual or reproductive health or HIV appointment	Significantly increased	See: (Dhakwa et al., 2021)	1
HIV testing	Significantly increased	See: (Nuwamanya et al., 2020)	1
HIV testing	No significant change	See: (Brody et al., 2022)	1
Adherence to HIV treatment	Significantly increased	See: (Hacking et al., 2019)	1
Adherence to HIV treatment	Mixed: significantly increased 2 of 6 studied reviewed; others found no significant increase	See: (Griffee et al., 2022)	1
Adherence to HIV treatment	No significant change	See: (Goldstein et al., 2023; Ivanova et al., 2019; MacCarthy et al., 2020)	3
Facility births	Significantly increased	See: (Thompson et al., 2019)	1
Health Outcomes			
HIV viral load suppression	No significant increase	See: (Hacking et al., 2019; Venter et al., 2019)	2
Pregnancy rates	No significant change	See: (Meherali et al., 2021; Smith et al., 2015)	2

The review found less evidence of medium- and longer-term impacts, such as behavior change and improvements in health outcomes resulting from a technology-enabled intervention. In terms of health behaviors, only one study showed a reduction in risky sexual behavior (Hernández-Torres et al., 2022), while three did not (Hu et al., 2023; Jamison et al., 2013; Lopez et al., 2014). One study reported increased condom use (Nuwamanya et al., 2020), while two others did not (Brody et al., 2022; Ybarra et al., 2013). Four studies found increased contraceptive use (Nuwamanya et al., 2020; Palmer et al., 2020; Smith et al., 2015; Unger et al., 2018), whereas one did not (Riley, 2014). For targeted health services, individual studies reported increases in keeping referrals to services (Dhakwa et al., 2021), testing for HIV (Nuwamanya et al., 2020), adherence to HIV treatment (Hacking et al., 2019), and facility births (Thompson et al., 2019). Conversely, one study found no significant change in HIV testing (Nuwamanya et al., 2020) and three studies found no significant change in adherence to HIV treatment (Goldstein et al., 2023; Ivanova et al., 2019; MacCarthy et al., 2020). Regarding health outcomes, two studies found no change in

HIV viral load suppression (Hacking et al., 2019; Venter et al., 2019), and two reported no change in pregnancy rates (Meherali et al., 2021; Smith et al., 2015).

Several factors must be considered when interpreting these results. First, the weak evidence is primarily due to the low methodological rigor of the published studies on the impact of these programs. It is possible, even likely, that important impacts exist but have not yet been identified. Several rigorous studies are underway, as indicated in published protocol papers. Moreover, international non-governmental organizations (INGOs) conducting potentially effective adolescent programming with technological interventions for SRH often lack the resources to perform rigorous impact studies. The rapidly evolving technology and diverse target populations (e.g., high risk of HIV, high risk of pregnancy) make it difficult to identify clear patterns of effective programs that can be generalized. The field is moving towards two-way interventions, and there is suggestive, albeit not definitive, evidence that these may improve outcomes (Griffee et al., 2022).

3.4. Adolescent Preferences (Research Question Two)

The literature on adolescent preferences for digital program delivery related to SRH is substantial and offers valuable insights, in contrast to the literature on program impacts, which often lacks methodological rigor. The preferences of adolescents are well documented across numerous studies. Of the 79 papers reviewed, 54 provided information on this topic, even those focused primarily on intervention types or impacts. Table 3 inventories positive features identified in the literature. The features are listed in order of the frequency with which they are mentioned in the citations.

Table 3. Positive features of reviewed SRH interventions for adolescents.

Feature	Citations	Number of Citations
Convenience of technology	See: (Brahme et al., 2020; Dev et al., 2019; Endehabtu et al., 2018; Feroz et al., 2021; Glik et al., 2016; K. L'Engle et al., 2017; Macharia et al., 2021; Nalwanga et al., 2021; Palmer et al., 2020; Vahdat et al., 2013; Wang et al., 2022; Winskell et al., 2018)	12
Improved understanding of SRH and improved behaviors (as reported in qualitative studies)	See: (Bergam et al., 2022; Dev et al., 2019; Feroz et al., 2021; McCarthy et al., 2020; Winskell et al., 2019)	5
Privacy	See: (Ivanova et al., 2019; K. L'Engle et al., 2017; Reynolds et al., 2019; Somefun et al., 2021)	4
Being involved in developing the intervention (e.g., game app)	See: (Ahonkhai et al., 2021; Reynolds et al., 2019; Ybarra et al., 2020)	3
Tailored content to target group	See: (Ames et al., 2019; Kharono et al., 2022; K. L. L'Engle et al., 2016)	3
Involvement of facilitators or mentors in two-way and group chats; maintaining regular contact with mentors	See: (Atujuna et al., 2021; Batista et al., 2020; Hacking et al., 2019)	3

Table 3. *Cont.*

Feature	Citations	Number of Citations
Broad information on the internet	See: (Ivanova et al., 2019; Mangone et al., 2016)	2
Simple language	See: (Bull et al., 2010)	1
Preference for existing apps such as WhatsApp or Facebook	See: (Henwood et al., 2016)	1
Providing access to health providers	See: (Huang et al., 2022)	1
Parental support/acceptability to parents	See: (Winskell et al., 2019)	1

The most frequently cited positive feature is the convenience of technology, mentioned in 12 studies (Brahme et al., 2020; Dev et al., 2019; Endehabtu et al., 2018; Feroz et al., 2021; Glik et al., 2016; K. L'Engle et al., 2017; Macharia et al., 2021; Nalwanga et al., 2021; Palmer et al., 2020; Vahdat et al., 2013; Wang et al., 2022; Winskell et al., 2019). Adolescents appreciate having cell phones, particularly smartphones, readily available throughout the day without restrictions. This convenience extends to mentors and parents involved in the studies. Improved knowledge and health behaviors resulting from interventions were reported in five qualitative studies, which often differ from impact study results, highlighting the importance of considering both types of evidence. Improved understanding of SRH is noted as a positive feature in five studies (Bergam et al., 2022; Dev et al., 2019; Feroz et al., 2021; MacCarthy et al., 2020; Winskell et al., 2019). Privacy is another highly valued positive feature (Ivanova et al., 2019; K. L'Engle et al., 2017; Reynolds et al., 2019; Somefun et al., 2021). Additionally, involvement in developing the app (Ahonkhai et al., 2021; Reynolds et al., 2019; Ybarra et al., 2020); tailored content for targeted groups (Ames et al., 2019; Kharono et al., 2022; K. L. L'Engle et al., 2016); the involvement of facilitators/mentors (Atujuna et al., 2021; Batista et al., 2020; Hacking et al., 2019); and broad accessibility to information on the internet (Ivanova et al., 2019; Mangone et al., 2016) are other significant positive features. Less frequently mentioned preferences include simple language, existing apps like WhatsApp, access to health providers, and acceptability by parents (Bull et al., 2010; Henwood et al., 2016; Huang et al., 2022; Winskell et al., 2018).

Despite these positive features, several potential problems with SRH technological interventions were identified (Table 4). The most prevalent issue, cited in nine studies, is the lack of access to cell phones, especially smartphones (Ames et al., 2019; Chukwu et al., 2021; Endehabtu et al., 2018; Goldstein et al., 2023; Greenleaf et al., 2019; Kharono et al., 2022; Laidlaw et al., 2017; Ochieng et al., 2022; Waldman & Stevens, 2015). This issue is more common among younger, female, and very low-income adolescents and persists in recent studies. Another problem mentioned in eight studies is that adolescents' interest in novel material may wane over time, leading to reduced attention to SRH information (Batista et al., 2020; Fatori et al., 2020; Gonsalves et al., 2019; Henwood et al., 2016; Huang et al., 2022; Jamison et al., 2013; Karusala et al., 2021; Ybarra et al., 2014). Low technological literacy, mentioned in six studies, is another significant issue, particularly among groups lacking access to technology (Feroz et al., 2021; Ibegbulam et al., 2018; Ivanova et al., 2019; Jamison et al., 2013; Mwaisaka et al., 2021; Ybarra et al., 2012). These adolescents may struggle with using smartphones or computers for internet searches, apps, or other tasks, which are often crucial for interventions targeting specific groups like females for contraceptive education.

Table 4. Problems for adolescents with SRH technological interventions.

Feature	Citations	Number of Citations
Low accessibility and/or high cost of cell phones/smartphones, especially for younger, female, and/or low-income adolescents	See: (Ames et al., 2019; Chukwu et al., 2021; Endehabtu et al., 2018; Goldstein et al., 2023; Greenleaf et al., 2019; Kharono et al., 2022; Laidlaw et al., 2017; Ochieng et al., 2022; Waldman et al., 2018)	9
Lack of interest in material and/or lowered interest over time; discontinuation of use of app/website	See: (Batista et al., 2020; Fatori et al., 2020; Gonsalves et al., 2019; Henwood et al., 2016; Huang et al., 2022; Jamison et al., 2013; Karusala et al., 2021; Ybarra et al., 2014)	8
Low technological literacy	See: (Feroz et al., 2021; Ibegbulam et al., 2018; Ivanova et al., 2019; Jamison et al., 2013; Mwaisaka et al., 2021; Ybarra et al., 2012)	6
Concerns about privacy, especially with borrowed phones	See: (Ames et al., 2019; Karusala et al., 2021; Mathenjwa et al., 2023; Somefun et al., 2021; Waldman et al., 2018)	5
Low linguistic competency	See: (Dev et al., 2019; Feroz et al., 2021; K. L. L'Engle et al., 2016)	3
Low access to/use of the internet; poor network coverage	See: (Adams et al., 2017; Feroz et al., 2021; Pfeiffer et al., 2014)	3
Technical glitches (e.g., in the app)	See: (Nuwamanya et al., 2020; Somefun et al., 2021; Venter et al., 2019)	3
Need for in-person engagement with health providers and counselors	See: (Batista et al., 2020; Regmi et al., 2022)	2
Potential parental concerns about misuse of phones (e.g., pornography); potential need for parental consent	See: (Laidlaw et al., 2017; Ochieng et al., 2022)	2

Concerns about a lack of privacy, mentioned in five studies, arise primarily when technology is shared among family and friends, posing a challenge despite privacy being seen as a positive feature (Ames et al., 2019; Karusala et al., 2021; Mathenjwa et al., 2023; Somefun et al., 2021; Waldman & Stevens, 2015). Other problems, though mentioned less frequently, include low linguistic competency (Dev et al., 2019; Feroz et al., 2021; K. L. L'Engle et al., 2016); poor internet access and network connections (Adams et al., 2017; Feroz et al., 2021; Pfeiffer et al., 2014); a lack of personal connection with mentors or providers (Batista et al., 2020; Regmi et al., 2022); technical glitches with the app (Nuwamanya et al., 2020); and parental concern about potential misuse of phone technology (Laidlaw et al., 2017; Ochieng et al., 2022). These issues underscore the need for careful consideration and design of SRH digital interventions to address the diverse needs and challenges faced by adolescents.

3.5. Types of Investments (Research Question Three)

The literature review on the level and types of investments organizations make to develop and deliver digital SRH behavior change interventions revealed limited information on implementation costs. Only three papers addressed the cost to implementers. The "MOTIF" (Mobile Technology for Improved Family Planning) program in Cambodia was a mobile phone–based support program added to standard post-abortion family planning care (Hill et al., 2020). Participants received six automated interactive voice messages and phone support from a counselor. Depending on their responses, additional reminder messages were provided for those who chose oral or injectable contraceptives. A cost-effectiveness study conducted as part of the program evaluation detailed the provider's cost of airtime for a two-way intervention at USD 4.49 per client per year and the cost of the special counselor for the intervention at USD 2.52 per client. These costs are specific to the technological features added to the existing program and provide insights into the potential financial implications for similar interventions.

Another mobile technology-based family planning program, Mobile for Reproductive Health (m4RH), sponsored by the US Agency for International Development (USAID) and scaled up in Tanzania, Uganda, and Rwanda, also underwent a cost study (Riley, 2014). The implementation of m4RH in Tanzania cost USD 203,475 for 125,320 unique users in 2014, equating to USD 1.62 per user. Over half of this cost was attributed to the per-text cost of USD 0.03, which the program absorbed at no cost to the user. Researchers explored the feasibility of shifting the text costs to users for sustainability but found that this would significantly reduce the program's reach, particularly to low-income groups, thereby challenging its sustainability.

A different sustainability approach was proposed by an SRH program in Uganda (Ybarra et al., 2020), which considered incorporating advertising into their mobile phone app. While the potential savings from allowing advertisements were not reported, this approach highlights alternative funding mechanisms that could support the financial sustainability of digital SRH interventions. However, the literature still lacks comprehensive data on the overall costs and economic models needed to sustain such programs, indicating a critical gap in understanding the full scope of investments required for effective and enduring digital health interventions for adolescents.

Researchers examining the MOTIF and m4RH programs did not enumerate the fixed costs of developing the interventions. However, this was addressed in a program in India that used voice or text messages to remind patients on HIV treatment to take their drugs (Rodrigues et al., 2014). This study tabulated both fixed costs, including those for message development and equipment, and variable per-patient costs. In 2013 US dollars, the fixed annual cost at the national level to develop and oversee the program was USD 6000; the fixed annual cost per treatment facility was USD 689; and the variable per-patient cost was USD 1.88 for voice messages and USD 0.78 for text messages.

All three programs for which costs were reported were text-based "push" programs implemented early in the review period. These programs relied on texts or voice messages sent to traditional cell phones, not apps on smartphones. For two-way programs, costs are likely higher, as developing an app from scratch or adapting an existing one would be costlier. An important consideration is the cost for the adolescent and/or their family. Several studies mentioned this issue, with one quantifying the weekly cost of mobile data for adolescents in Kenya as an average of USD 1.20 per week in 2019 (Karusala et al., 2021).

Additionally, the cost of a cell phone or smartphone was not documented in any of the studies reviewed. The cost of owning and maintaining a cell phone, particularly a smartphone, is a critical factor when implementing an intervention based on the assumption that an adolescent will own or have access to one. When phones are shared within a family,

privacy issues can create additional barriers to participation. Focus groups revealed that adolescents and their parents suggested subsidizing the costs of phones and phone usage to enable low-income adolescents to participate in these programs. Subsidizing these costs will add to the programmatic expenses outlined above.

4. Discussion

This detailed review of qualitative and quantitative studies highlights several key findings regarding the use of technology-based interventions for adolescent sexual and reproductive health in low- and middle-income countries. Adolescents show a strong preference for using technology, finding it a convenient medium for accessing SRH information. Various technology-based interventions have been implemented, offering diverse SRH information and services to different target populations, including adolescents living with HIV, pregnant/postpartum adolescents, and students. Early interventions primarily utilized basic cell phones, while recent programs increasingly leverage smartphones, enabling more interactive communication. Involving adolescents and stakeholders in developing these interventions is crucial, emphasizing the importance of pre-testing materials and incorporating adolescent feedback to ensure they remain acceptable and engaging.

Despite the positive reception and engagement, several barriers persist, particularly due to technology costs, which limit participation among adolescent females, younger adolescents, and marginalized, low-income groups. Qualitative data suggest that adolescents engage with and utilize SRH information through technology, but their attention spans require innovative and engaging content. This could mean changing the delivery modalities to include engaging ways of interacting, for example, through games or chatbots. It could also mean updating the content itself to ensure it remains current, inclusive, and in line with changing social norms. Furthermore, while privacy is generally a beneficial aspect of technology use for SRH, it becomes problematic when devices are shared with family or friends. Adolescents appreciate interactions with peers and mentors, as they can be deemed more approachable or less formal, in addition to receiving information through digital platforms.

The impact of technology-based SRH interventions on knowledge is generally positive, with most studies indicating improvements in SRH knowledge. However, results on their effects on health behaviors and service utilization are mixed, and there is no conclusive evidence that these interventions positively impact health outcomes. The synthesis of impact results is complicated by program diversity, rapidly evolving technology, and the extensive time and costs required for rigorous studies. No single approach has emerged as the definitive method for improving adolescent SRH outcomes through technology.

This review has several limitations. First, it is based only on published literature (including gray literature). As mentioned elsewhere, small nonprofit organizations often have no resources to conduct rigorous studies of effectiveness or, further, to publish such studies. We based the review only on English-language literature and used only two broad search engines. Consequently, some important studies may have been missed. Finally, it does not include research that is in progress or research that had not yet been published at the time of the review in 2023. This review also has numerous strengths. It synthesizes the findings of 79 high-quality studies into three categories of interventions for easy assessment. It also succinctly reviews the impact evidence of these studies and groups them by positive and negative features according to end users so implementing organizations can easily replicate intervention successes and avoid common pitfalls.

Overall, there is a critical need for more robust impact studies with stronger designs and larger sample sizes to determine the effectiveness of technology-based SRH programs on adolescent health. Limited cost data suggest that these programs are relatively low-

cost compared to intensive in-person SRH education programs, but more research is needed, particularly for recent two-way programs. The sustainability of donor-funded pilot programs remains challenging, with no reviewed programs achieving self-funding. This review underscores the potential of technology-based interventions to enhance adolescent SRH knowledge while highlighting significant gaps in evidence regarding their impact on health behaviors and outcomes. Comprehensive, large-scale, and methodologically rigorous studies are urgently needed to inform the design and implementation of effective technology-based SRH programs for adolescents. Finally, additional reviews should be completed on an annual or semi-annual basis due to rapid advancements in the field, particularly around the use of artificial intelligence. Continued synthesis of the literature will benefit public health implementing agencies and ensure users are receiving high-quality interventions based on the most up-to-date evidence.

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